

Due Mon/21-Oct-2024

HW Instructions:

1. You are to **develop**, **solve** and **document** your solution:
 - Abaqus Model (.cae file), as demonstrated in class **AND**
 - MATLAB Notebook (.mlx file), as demonstrated in class **AND/OR**,
 - Complete the homework set description (this document) with your solution.
2. Save your solution files adding your Last and First name to the end of the filename:
 - for example: **me5634_hw4_LastName_FirstName**.
 - Abaqus Model (.cae) as: **me5634_hw4_LastName_FirstName.cae**
 - MATLAB Notebook (.mlx) as: **me5634_hw4_LastName_FirstName.mlx** and/or
 - Convert this homework set description (this document), into a .pdf file as:
me5634_hw4_LastName_FirstName.pdf
3. Please upload the completed solution files to the Canvas Quiz for the assignment.
 - Abaqus Model (.cae) file
 - MATLAB Notebook (.mlx) **and/or**
 - Homework set description solution (.pdf)
4. Please refer to your MATLAB Notebook and/or handwritten solutions while taking the quiz. Your homework solutions will be required to answer questions in the quiz used to grade the assignment.

Note:

- Logic, formulas and calculations leading to answers must be **documented**.
- Any points or lines used from graphs must be marked and labeled, for example, stress concentrations figures.
- Numerical answers should have **sufficient significant figures**, and **correct units** are required for full credit.
- I believe there is merit in developing an engineering solution so you can revisit and learn from your work. It is also possible that you may revise your solution to apply a best practice, correct a solution, or prepare for an exam.

Honor Code Instructions:

1. The Honor Code is being applied to these HW/Quiz solutions.
2. There is merit in working and learning together. You may discuss the problems with other students in class, however, you are responsible for your own work.
3. Copying someone's solution is **NOT** acceptable, it is a violation of the Honor Code.
4. Please sign/date the VT Honor Code Pledge in the MATLAB Notebook:

Virginia Tech Honor Code Pledge:

I pledge that I have neither given nor received unauthorized assistance on this assignment:

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A stepped shaft is shown is loaded by two spur gears and supported by two radial bearings at the left and right ends of the shaft giving bearing reactions, R_A and R_B . The shaft is machined from UNS G41400 Cold-Drawn steel with the following properties, $E = 207$ GPa, and Yield Strength, $S_y = 620$ MPa. The shaft has an allowable stress, $\sigma_{\text{allowable}} = 600$ MPa and allowable deflection, $\delta_{\text{allowable}} = 0.2$ mm. “Defeature” the shaft by ignoring the shoulders on the shaft for the bearings at A and B.

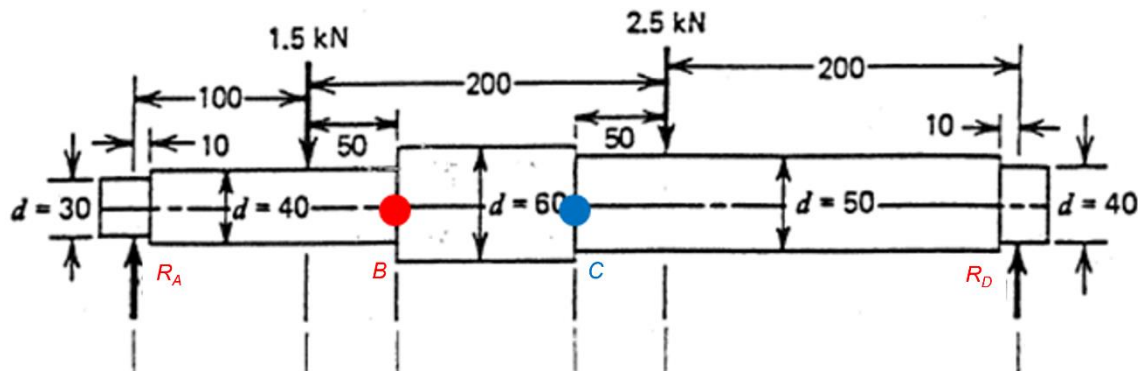


Figure 1: Stepped Shaft

1. (10) Perform a Design Hand Calculation for an independent check of the finite element solution.
 - a. Use superposition of Shigley’s Table A-9 Case 6 to estimate the deflection.
 - i. Lower Bound; use a single diameter (no step), $d = 50$ mm
 1. Tip Deflection: under the 1.5 kN and 2.5 kN loads,
 2. Rotation at left bearing, A, and right bearing, B.
 - ii. Upper Bound; use a single diameter (no step), $d = 40$ mm
 1. Tip Deflection: under the 1.5 kN and 2.5 kN loads,
 2. Rotation at left bearing, A, and right bearing, B.
 - b. Plot the Load, Shear and Moment diagram for the shaft.
 - c. Where is the critical section, what is the nominal and actual stress at the critical section; does the shaft “fail” in stress/deflection?

Hint: Write simple MATLAB m-files to compute the displacement and slope for Shigley’s Table A-9: Case 6; Simple Supports-Intermediate Load. Suggestion, write simple functions, such as:

```
[A, I, J, Z] = me5634_SectionCircularSolid(d)
[Uy, θz] = me5634_ShigleyTableA9Case6(ei, f, x, l, a)
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2. (10) Using the 1D Beam Element developed in class (Cook pg. 26, Eqn 2.3-6), develop a MATLAB-Based FE solution using the Partitioned Form of the Direct Stiffness Method. Develop the FE code from MATLAB examples in class.
 - a. Compute the element stiffness matrix for each element.
 - b. Assemble the global stiffness matrix using the partitioned matrix approach.
 - c. Solve the displacement equation for the unknown displacements and the reaction equation for bearing reactions.
 - d. Compute the maximum internal bending stress in each element
 - e. Using the elastic curve, PLOT the following field quantities:
 - i. Transverse Displacement, Rotation, Moment, and Transverse Shear
3. (10) Solve the stepped post problem in Abaqus using the Euler-Bernoulli cubic beam element (B23) and the Shear Flexible Element (B22).

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Nodal Solution								
Node	Displ [mm]	Displ. [mm]	Transverse Displacement [mm]			Rotation [radians]		
	d = 40 Design	d = 50 Design	Hand/MATLAB	Abaqus B23	Abaqus B22	Hand/MATLAB	Abaqus B23	Abaqus B22
N1								
N2								
N3								
N5								
N5								
N6								

Elemental Solution							
Element	Bending Stress	Bending Stress Top Fiber		Bending Stress Top Fiber		Bending Stress Top Fiber	
	Design [MPa]	Hand/MATLAB [MPa]		Abaqus B23 [MPa]		Abaqus B22 [MPa]	
		Node i	Node j	Node i	Node j	Node i	Node j
E1							
E2							
E3							
E4							
E5							

Design Factor of Safety: Deflection and Yield				
	Design	Hand/MATLAB	Abaqus B23	Abaqus B22
$n_{\text{Deflection}}$				
n_{Yield}				